

Job Interview in the AI-Era: Coding, Systems, Agents

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Lecture 4

ML:Data pipeline

Onboarding a project

Preparing coding interview

Outline

- Preparing for coding interview
- Onboarding a new project
- ML Data Pipeline

Preparing For Coding Interview(without AI)

- Assumptions: you're doing coding interview without access to any GenAI during interview.
- More on basics:
 - time limit and complexity: 1s, input size->complexity, rough estimate on $\log$, $\exp$
 - debugging:
 - stress test
 - debugger: pdb
 - print, some mental models
 - example: writing a correct binary search:
 - $\text{mid} = (l+r)/2$, $hi < -\text{mid}$ / $lo < -\text{mid}$
 - `int l, r?`
 - loop invariants, brute force for small cases
 - floating point numbers

Preparing For Coding Interview(without AI)

- How to prepare:
- solve more problems
 - if you have solved 200+ problems on leetcode/any platform you can skip this part, the purpose of this section is to probably help you save some time
 - difficulty comes from 2 parts:
 - problem solving, we will be mostly talking about how to effectively train this
 - coding, also look at this: <https://www.deep-ml.com/>
- Principles on problem solving:
 - Scaling law: <http://www.incompleteideas.net/Incldeas/BitterLesson.html>
 - Basic idea for problem solving: a problem is like a point in a high dimensional space, solving it is like reaching that point in the search space
 - In general, there is no short cut, but you should avoid ineffective repetition
 - The only important thing is to, build your own general problem solving framework that can effectively explore a large 'space'

Preparing For Coding Interview(without AI)

- Building your toolkit:
 - Search Instead of random trials
 - Know what you can do during the process
 - Try small examples
 - Do observations
 - Try pattern matching
 - How to do branching:
 - Problem solving paradigms:
 - Brute force, Greedy, Divide and Conquer, Binary search, Graph, Data Structures...
 - Transform the problem
 - look at it in different ways
 - Use your hunch
 - Write it down

Preparing For Coding Interview(without AI)

- Exercises:
 - <https://open.kattis.com/contests/m4aqwi/problems/exchangerates>
 - Hint: floating point errors
- <https://leetcode.com/problems/maximum-sum-of-3-non-overlapping-subarrays/description/>
 - try difference routes

Preparing For Coding Interview(without AI)

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AI assisted onboarding

- Why it's relevant
- principles of onboarding without AI
 - some workflow:
 - overview: readme/docs
 - understand architecture: core components, dataflow...
 - setup and run
 - start with concrete small tasks
 - use debugging tools to learn
 - it's different from what you've learned from class

AI assisted onboarding

- With AI
 - Read this: <https://newsletter.eng-leadership.com/p/how-to-use-ai-to-onboard-into-a-codebase>
 - what's the difference? friction, blackbox
- Some ideas
- Exercises:
 - Add permission pop ups to pi coding agent, don't ask an LLM for how to do it(it might be simpler than how you plan to do it)
 - suppose you are provided 2 sets of api:
 - one to call LLM and get response
 - one to call one of these 4 tools: “read_file, write_file, edit, run_command”
Write a program such that it can run as a minimal coding agent
 - explain pi's core agent loop, ask a LLM to give you some questions regarding design/architecture of this project and try to answer it

ML: data pipelines

- just read the references if you're interested:
- Lecture 4: Data Management - The Full Stack
- Data Distribution Shifts and Monitoring
- CS 329S | Syllabus

Other References:

- * stress testing for coding interviews: <https://codeforces.com/blog/entry/81841>
- How to solve it:
https://www.hlevkin.com/hlevkin/90MathPhysBioBooks/Math/Polya/George_Polya_How_To_Solve_It_.pdf
- AI assisted onboarding:
<https://medium.com/@ramtop/ai-assisted-onboarding-5de547520b76>